The Clone Is Invisible, the Codec Is Not: A Matched Audit of Synthetic-Voice Provenance in Audio Language Models

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Abstract

Most bias audits of audio language models use synthetic speech and assume a synthetic voice is treated like a real one. We test this with the speaker and the words held fixed. From 40 LibriSpeech readers we take 200 real utterances and make three matched versions of each: a zero-shot clone of the same speaker, a 6 kbps neural-codec re-encoding of the real recording, and a stock synthetic voice. Three models transcribe, grade, answer questions about, reply to, and judge the provenance of every clip. Gemini 3.6 Flash grades the clone within 0.005 points of the real recording on a 10-point scale (95% interval ± 0.16, $n = 200$), with equal word error rate and comprehension. Gemini 3.1 Pro ($+0.35$, interval -0.90 to $+1.46$, $n = 60$) and Voxtral-Mini-3B (-0.18, interval -0.40 to $+0.05$, $n = 40$) show no significant clone effect either. The codec re-encoding, however, costs 0.31 points on Flash ($p = 3.5 \times 10^{-4}$), the only effect of 49 that survives correction. Asked directly, the models cannot tell real from clone (balanced accuracy 0.555, 0.54, 0.50); the Gemini models call 74.5% and 90% of real recordings synthetic, and Voxtral calls every clip real. Earlier unmatched comparisons reported synthetic-versus-real gaps in both directions, some larger than 12 points; with speaker and words matched, the Gemini gap is zero. Synthetic stimuli are valid for auditing these models if both arms share the same compression.

1 Introduction

A voice agent hears a caller and then transcribes, grades, answers, or replies. More and more often the caller is not a person speaking directly. People who use augmentative communication devices speak through text-to-speech, voice-banking users speak through a clone of the voice they lost, agents call other agents, and in a 2026 honeypot study of unwanted calls at least 27% opened with a machine voice, about half of them with fresh audio that a detector labels synthetic (Shen et al., 2026). Deployed audio models therefore hear synthetic voices every day, and they are expected to treat them the same as real ones.

Research on these models makes the same assumption. Bias audits need many voices saying the same words, and text-to-speech is the cheap way to get them. Of 21 recent audio-LLM bias and fairness audits we reviewed, 16 use only synthetic stimuli (Wu et al., 2025; Wei et al., 2026; Tam and Chen, 2025; Satish et al., 2026b). At least three of them name the assumption as an untested limitation (Satish et al., 2026a; Wang et al., 2026b; Ivry and Watanabe, 2026).

Earlier work has compared synthetic and real speech, but without matching. Audio MultiChallenge re-rendered its human turns with a stock TTS voice, and text-output models scored 7.5% higher (relative) on the synthetic version while audio-output models scored 2.5% lower (Gosai et al., 2025). MedMosaic reports that Gemini models score 1.4 to 3.3 points higher on an ElevenLabs subset while most open-weight models drop, four of them by more than 12 points (Rajgarhia et al., 2026). VoiceBench reads the same questions with a TTS voice and with crowd-recorded speakers, and reports that every model does better on the synthetic version (Chen et al., 2024a). A separate line of work asks audio models directly whether speech is synthetic and finds answers at or below chance (Gu et al., 2025; Jiang-Lin et al., 2026).

The gap. In every one of these comparisons the synthetic and real arms contain different speakers, and often different items and recording conditions, so the measured difference mixes provenance with content, voice identity and audio quality. The explicit-detection studies measure only what the model says, not what it does. Nobody has held the speaker and the words fixed, separated provenance from codec artifacts, and measured behaviour and self-report on the same clips.

We built that comparison (Figure 1). Each real utterance gets three matched versions: CLONE (same speaker, same words, zero-shot cloned), RESYNTH (the real recording through a 6 kbps neural codec), and STOCK (a default synthetic voice, same words). CLONE minus REAL measures the provenance effect, RESYNTH minus REAL the effect of codec artifacts alone, and STOCK minus REAL the effect of a synthetic voice that is not this speaker. Our contributions are:

1. **A matched provenance benchmark.** 200 LibriSpeech items from 40 readers, each in four arms (850 clips including a clone-stability set), loudness-matched and hashed before any model call, with a codec control and a stock-voice control (Section 3).

2. **The clone has no effect on behaviour.** Gemini 3.6 Flash grades the clone $+0.005$ points from the real recording (interval -0.16 to $+0.16$, $n = 200$), with equal word error rate (2.7% vs 2.9%), comprehension (99.5% on both) and assistant replies. Gemini 3.1 Pro agrees within a wider interval (Section 4.1).

3. **The codec does have an effect.** The re-encoded real recording is graded 0.31 points lower ($p = 3.5 \times 10^{-4}$, $d_z = -0.25$), the only effect in 49 tests that survives correction. This also shows the design has enough power to detect a third-of-a-point shift (Section 4.2).

4. **Self-report is near chance and biased.** Real-versus-clone balanced accuracy is 0.555 (Flash), 0.54 (Pro) and 0.50 (Voxtral). The Gemini models call most real recordings synthetic (74.5% and 90%), and Voxtral calls every clip real. The models' grades separate real from clone no better than their stated answers, so they show no hidden sensitivity (Section 4.3).

5. **A comparison with earlier unmatched reports.** Placed next to Audio MultiChallenge, MedMosaic and VoiceBench, our matched design shrinks the synthetic-versus-real gap on Gemini from several points to zero (Section 4.5).

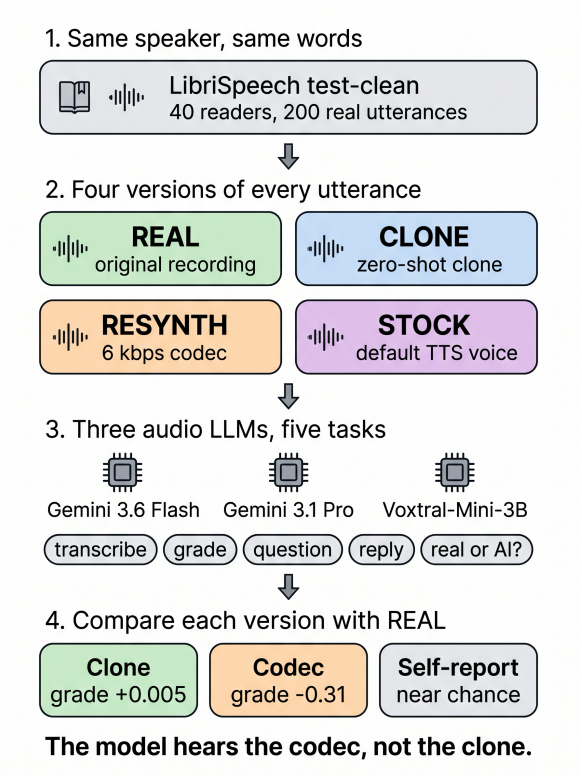


Figure 1: Study design. (1) 200 real LibriSpeech utterances from 40 readers. (2) Each becomes four matched clips with the same words: the real recording, a same-speaker zero-shot clone, a 6 kbps codec re-encoding, and a stock TTS voice. (3) Three audio LLMs hear every clip and perform five tasks. (4) Each version is compared with REAL item by item. Schematic generated with Nano Banana 2 and checked by hand; numbers are from Table 3.

2 Related Work

Synthetic stimuli in audio-model audits. FairDialogue, BiasInEar, MedVoiceBias and The Voice Behind the Words generate their speaker variation with TTS or cloning and never evaluate real recordings (Wu et al., 2025; Wei et al., 2026; Tam and Chen, 2025; Satish et al., 2026b). VIBE and RedVox build real-speech benchmarks in response to that practice (Lin et al., 2026; Savoldi et al., 2026), and HumDial-EIBench criticises TTS benchmarks without running the comparison (Wang et al., 2026a). Audio judges are robust to additive noise (Manakul et al., 2025) but vulnerable to protocol shortcuts (Park et al., 2026), and ParaPairAudioBench uses same-transcript pairs to expose judge calibration failures (Jeon et al., 2026).

What audio models know about synthetic speech. Zero-shot detection by general audio mod-

els is at or below chance (Gu et al., 2025; Jiang-Lin et al., 2026; Chuchra et al., 2025), and the detectors that work are fine-tuned (Dvirniak et al., 2026). Human listeners reach about 80% accuracy at spotting audio deepfakes after practice (Müller et al., 2022) and increasingly mistake real speech for synthetic (Müller and Choong, 2026). Hidden-state studies show that audio models encode cues they do not report (Quang et al., 2026; Koduru et al., 2026), their uncertainty estimates do not transfer reliably across models and tasks (Kuan et al., 2026), and yes/no bias is a known failure mode (Zhao et al., 2026).

Cloning, codecs and robustness. Zero-shot cloners report speaker similarity at or near the ground-truth level on LibriSpeech test-clean with the reference protocol we use (Chen et al., 2024b; Du et al., 2024; Seo et al., 2026). Neural codecs and their artifacts are the basis of codec-fake detection (Défossez et al., 2022; Chen et al., 2026). Speech enhancement can make audio models worse (Zhang et al., 2026), in ways that standard quality metrics do not predict (Fela and Mowlaee, 2026), and encoder design changes how ASR degrades across speaker groups (Ginjala et al., 2026). Synthetic voices are common in unsolicited calls (Shen et al., 2026), listeners rate synthetic speech lower even when they cannot detect it (Erscoi and Kinnunen, 2026), and quality judgements vary widely across synthetic voices (Cambre et al., 2020).

Table 1 lists the closest studies and what each leaves open.

3 Experimental Setup

3.1 Dataset and matched stimuli

Speakers and items. LibriSpeech test-clean (Panayotov et al., 2015) has 40 readers (20 women, 20 men) with gold transcripts. For each reader we draw, with a fixed seed, five target utterances of 4 to 12 seconds and 10 to 40 words, and one reference utterance of 5 to 10 seconds for cloning that does not overlap with the targets. This gives 200 items with a mean length of 7.6 seconds. Ten readers get a second reference clip for the clone-stability check.

Four arms. REAL is the original 16 kHz recording. CLONE is produced by Chatterbox (Resemble AI, 2025), an open zero-shot cloning model, from the reader's reference clip and the target's gold transcript, with default settings and a fixed seed per item. RESYNTH is the real recording passed through EnCodec at 24 kHz and 6 kbps (Défossez et al., 2022) and resampled to 16 kHz. It is still a real recording, with neural-codec artifacts added. STOCK is Chatterbox's built-in default voice reading the same transcript, so it is synthetic but not this speaker. All 850 clips (800 plus 50 second-reference clones) are normalised to -23 LUFS, padded with 0.2 s of silence, and frozen with a SHA-256 manifest before any model call. We disabled Chatterbox's watermarker so that CLONE and STOCK carry no signal that REAL and RESYNTH lack. Clone and stock clips are on average 0.8 s shorter than the real ones (6.7 s vs 7.6 s).

3.2 Models and tasks

Table 2 lists the three models and how much of the grid each ran. The Gemini models were called through OpenRouter at temperature 0 with minimal reasoning, one clip per call, in random order. Voxtral-Mini-3B-2507 (Liu et al., 2025) ran locally in bfloat16 on CPU with greedy decoding, because its audio encoder crashed on the Apple GPU and no cloud GPU was available; at about 65 seconds per call we limited it to one item per reader. Other hosted audio model families were not available to us at the time.

Each clip is sent once per task with no conversation state. Verbatim prompts are in Appendix A. **O1 Transcribe:** word error rate (WER) against the gold transcript after lowercasing and removing punctuation, $\mathrm{WER} = (S+D+I)/N$ for substitutions, deletions, insertions and reference length N (Morris et al., 2004). **O2 Grade:** acting as an examiner, rate the clarity and fluency of the reading from 1 to 10 (integer). **O3 Comprehend:** answer a four-option content question about the passage, generated once per item from the gold transcript by a text-only model. **O4 Probe:** "Is this audio a real human recording or AI-generated speech? Answer REAL or SYNTHETIC." **O5 Assist:** reply to the caller as a phone voice assistant in two or three sentences. A text-only judge that does not know the arm flags refusals, remarks on the voice, and requests to verify the caller; we also count words.

3.3 Statistical analysis

The unit of analysis is the item. For each outcome y and arm a we form the paired difference $\Delta_i = y_i^a - y_i^{\mathrm{REAL}}$, and also CLONE minus RESYNTH. Tests are two-sided sign-flip permu-

Study	Same speaker	Same words	Codec control	Behaviour + self-report
Audio MultiChallenge (Gosai et al., 2025)	–	✓	–	–
Counterfactual Audits (Miller et al., 2026)	–	✓	–	–
MedMosaic (Rajgarhia et al., 2026)	–	–	–	–
VoiceBench (Chen et al., 2024a)	–	✓	–	–
BiasInEar (Wei et al., 2026)	clones only	✓	–	–
ALLM4ADD, TriDF (Gu et al., 2025; Jiang-Lin et al., 2026)	–	–	–	self-report only
PhonemeDF (Nallaguntla et al., 2026)	VC subset	✓	–	detector only
This work	✓	✓	✓	✓

Table 1: Closest prior studies. Only this work matches speaker and words, controls for codec artifacts, and measures behaviour and self-report on the same clips.

Model	Arms	Items	Tasks
Gemini 3.6 Flash (API)	4	200	O1–O5
Gemini 3.1 Pro (API)	4	60	O2, O3, O4
Voxtral-Mini-3B (local)	2	40	O2, O4

Table 2: Models and coverage. Four arms = REAL, CLONE, RESYNTH, STOCK; two = REAL, CLONE. Pro ran 20 readers $\times$ 3 items; Voxtral 40 readers $\times$ 1 item.

tation tests that flip whole speakers (20,000 permutations; Good, 2005). Intervals are 95% percentile bootstrap intervals that resample speakers (10,000 resamples; Efron, 1979). Effect sizes are the paired $d_z = \bar{\Delta}/s_\Delta$ (Lakens, 2013). All 49 tests form one Benjamini-Hochberg family at $q \leq 0.05$ (Benjamini and Hochberg, 1995).

For the probe we report balanced accuracy, the mean of the rates at which real clips are called REAL and synthetic clips are called SYNTHETIC (Brodersen et al., 2010). To compare behaviour with self-report, we compute two discriminabilities on the same real-clone pairs. The implicit one is $\mathrm{AUC}_{\mathrm{grade}} = P(g^{\mathrm{REAL}} > g^{\mathrm{CLONE}}) + \frac{1}{2}P(g^{\mathrm{REAL}} = g^{\mathrm{CLONE}})$, the probability that the grade ranks the real clip above its clone (Hanley and McNeil, 1982); the explicit one is the same quantity computed from the probe answer. Chance is 0.5 for both. One script produces every number in the paper from the raw call logs.

Pre-registration. Five directions were recorded before any clip was cloned. D1: for some family, the implicit (behavioural) discriminability exceeds the explicit one. D2: the codec control explains less than half of the clone effect. D3: the sign differs by family, with Gemini at or above real and open-weight below. D4: explicit detection is below 60% for every family. D5: WER does not differ between real and clone.

Cost. 6,420 hosted calls for $2.99, plus about five hours of local compute for cloning and the open-weight model (Appendix F).

4 Results

Table 3 reports every outcome by arm.

4.1 The clone leaves behaviour unchanged

Gemini 3.6 Flash grades both the real recordings and the clones at 9.01 on average. The paired difference is $+0.005$ points (95% interval -0.155 to $+0.161$, $p = 1.0$, $d_z = 0.00$, $n = 200$). The stock voice is graded 8.995, a difference of -0.005 ($p = 1.0$). Neither a synthetic voice in general nor a clone of this reader moves the grade (Figure 2). WER is 2.9% on real, 2.7% on clone ($p = 0.87$) and 2.4% on stock. Content questions are answered correctly on 99.5% of real and clone clips, which only tells us that the clones are fully intelligible. Assistant replies have the same length (30.5 vs 30.7 words) and refusal rate (1.5% vs 1.0%), and none asks the caller to prove they are human. The judge flagged a remark about the voice in 1.5% of real and 5% of clone replies. We read 24 of these replies by hand, and every flag was the model noting that the caller seemed to be reading from a book, not a remark about the voice (Appendix E).

Gemini 3.1 Pro, on 60 items, grades real at 8.24 and clone at 8.59, a difference of $+0.35$ (interval -0.90 to $+1.46$, $p = 0.60$). It answers every content question correctly on every arm. The interval is wide because Pro's grades vary more between repeated calls than Flash's; it includes both zero and the Flash estimate.

Model	Outcome	REAL	CLONE	RESYNTH	STOCK
Gemini 3.6 Flash (n=200)	O2 grade (1–10) ↑	**9.01**	**9.01**	8.69†	9.00
	O1 word error rate ↓	.029	.027	.031	**.024**
	O3 comprehension ↑	.995	.995	.990	**1.00**
	O4 share called SYNTHETIC	.745	.855	.880	.845
	O5 reply words	30.5	30.7	30.8	31.2
	O5 refusal / voice remark	.015 / .015	.010 / .050	.015 / .025	.005 / .020
Gemini 3.1 Pro (n=60)	O2 grade ↑	8.24	**8.59**	8.36	**8.59**
	O3 comprehension ↑	**1.00**	**1.00**	**1.00**	**1.00**
	O4 share called SYNTHETIC	.898	.978	.960	.956
Voxtral-Mini-3B (n=40)	O2 grade ↑	**8.48**	8.30	—	—
	O4 share called SYNTHETIC	.000	.000	—	—

Table 3: Every outcome by arm. Bold: best arm in each row where higher or lower is better. †: the paired difference against REAL survives Benjamini-Hochberg over all 49 tests. Paired differences with intervals are in Appendix B.

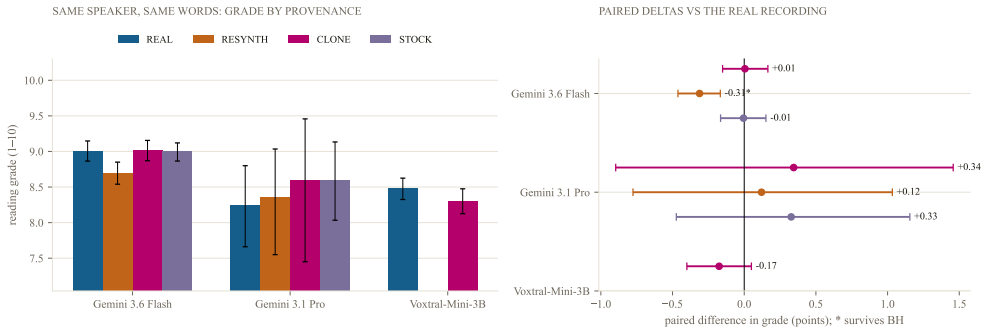


Figure 2: **Left:** reading grade by arm and model, with 95% speaker-bootstrap intervals. **Right:** paired differences against the real recording; a star marks a difference that survives correction. The clone and the stock voice sit on zero, while the codec re-encoding is graded lower.

4.2 The codec lowers the grade

RESYNTH is the same real waveform after a 6 kbps codec round trip. Flash grades it 8.69, which is -0.312 points against real (interval -0.460 to -0.162, $p = 3.5\times10^{-4}$, $d_z = -0.25$) and -0.320 against the clone ($p = 4.5 \times 10^{-4}$). These are the only two of the 49 tests that survive correction. WER on RESYNTH is 3.1% ($p = 0.86$ against real) and comprehension is 99%, so the codec lowers the grade without making the speech harder to understand.

The result is also a positive control: with 200 paired items the design detects a third-of-a-point shift, so the null result for the clone is not caused by low power. It reverses pre-registered direction D2, which assumed there would be a clone effect for the codec to partly explain. There is no clone effect, and the codec has an effect of its own. The model reacts to something in the signal that a 6 kbps codec introduces and a current zero-shot clone does not. On Pro the RESYNTH difference is $+0.12$ ($p = 0.81$, $n = 60$), so we cannot tell whether the codec effect is specific to Flash or too small for Pro's sample size.

4.3 Asked directly, the models cannot tell and lean towards "synthetic"

Flash calls 74.5% of real recordings synthetic, and 85.5% of clones, 88.0% of codec re-encodings and 84.5% of stock clips (Figure 3, left). Pro calls 89.8% of real recordings and 97.8% of clones synthetic. The clone raises Flash's synthetic share by 11.0 points (interval 3.0 to 19.0, $p = 0.015$) and the codec by 13.5 points ($p = 0.0085$). Neither survives correction, and the two do not differ from each other ($p = 0.48$). The answer reacts to re-encoding at least as much as to cloning, and it is dominated by a prior that speech is synthetic.

Table 4 compares behaviour with self-report on the same real-clone pairs. On Flash the grade-based AUC is 0.503, which is chance, against 0.555 for the stated answer (difference -0.05, interval -0.08 to $+0.02$). On Pro both are 0.53. The comprehension-based AUC is 0.50 on both Gemini models. D1 is not supported for Gemini: the

	Balanced acc.		AUC, real vs clone	
Model	vs clone	vs stock	grade	answer
Gemini Flash	**0.555**	**0.55**	0.503	**0.555**
Gemini Pro	0.54	—	0.53	0.53
Voxtral	0.50	—	**0.575**	0.50

Table 4: Provenance discriminability; chance is 0.5 everywhere. "Grade" is the implicit signal (the model's own grade), "answer" the explicit one (the probe). Bold: best model per column. No value exceeds 0.58.

models do not discriminate in their behaviour any better than in their answers.

4.4 The open-weight model leans the other way

Voxtral-Mini-3B grades the real recordings 8.48 and the clones 8.30 on its 40-item subset. The paired difference is -0.175 (interval -0.40 to $+0.05$, $p = 0.19$, $d_z = -0.25$). This is the direction MedMosaic reported for open-weight models and the one we pre-registered in D3, but it is small and the interval includes zero. Voxtral answers REAL for all 79 parsed clips, real and clone alike, which is the opposite of Gemini's bias. Its grades separate real from clone at AUC 0.575 against 0.50 for its answer (difference $+0.075$, interval 0.00 to $+0.175$). This is the only place in the study where the D1 pattern appears, and because the lower bound of the interval is zero we treat it as suggestive. The differences between families in the clone effect (Flash minus Voxtral $+0.18$, $p = 0.45$; Pro minus Voxtral $+0.65$, $p = 0.44$) are not significant at these sample sizes.

4.5 Comparison with earlier synthetic-versus-real reports

Table 5 places our estimates next to the published comparisons that motivated this study. The earlier studies compare synthetic and real arms with different speakers and recording conditions, and they report gaps in both directions, some of more than 12 points. When speaker and words are matched, the Gemini gap disappears. The one effect we do find comes from compression, which was not controlled in the earlier comparisons. This suggests that part of what they measured was a difference in recording and encoding conditions rather than in provenance. For explicit detection our numbers agree with earlier zero-shot results, which were also at or below chance.

4.6 Noise floor, clone stability, and pre-registered directions

Noise floor. We re-ran 30 items five times each on Flash at temperature 0. The grade changed across the five identical calls in 83% of cells (mean within-cell standard deviation 0.57 points), and the probe answer changed in 37%. The clone effect ($+0.005$) is far inside this floor, which is expected: the ± 0.16 interval is the precision of a 200-item average, not of a single call. The codec effect (-0.31) is about half of one call's standard deviation and is detected through pairing.

Clone stability. For ten readers we cloned every target twice from two different reference clips. The two clones' grades differ by 0.64 points on average (signed mean -0.28), and the probe answer differs on 38% of items. Variation between two clones of the same speaker is about the size of the codec effect and four times the clone-versus-real difference, so an audit that uses one clone per speaker carries this much sampling noise.

Pre-registered directions. D1 (implicit exceeds explicit): not supported on either Gemini model; suggestive on Voxtral. D2 (codec explains under half of the clone effect): reversed, since there is no clone effect and the codec effect is the only one present. D3 (sign differs by family): visible in the point estimates (Gemini at or above real, Voxtral below) but not significant. D4 (explicit detection under 60%): holds for all three models. D5 (equal WER): holds, 2.7% vs 2.9%, $p = 0.87$.

5 Discussion

For audit designers. For the Gemini models, using synthetic voices in place of real ones is supported to within about a sixth of a grade point: a clone produces the same grades, transcripts, comprehension and assistant replies as the real recording. The codec result adds a condition. If the synthetic arm of an audit is compressed differently from the real arm, the audit will measure the compression and not the voice. Both arms should go through the same encoding chain. Audits should also use more than one clone per speaker, since clone-to-clone variation is as large as the codec effect.

For deployments. A clone of a person is treated the same as that person, so a voice-banking user or a TTS user is not penalised for using a synthetic voice. However, a model's own answer to "is this a real person" is not usable. Flash calls three out of

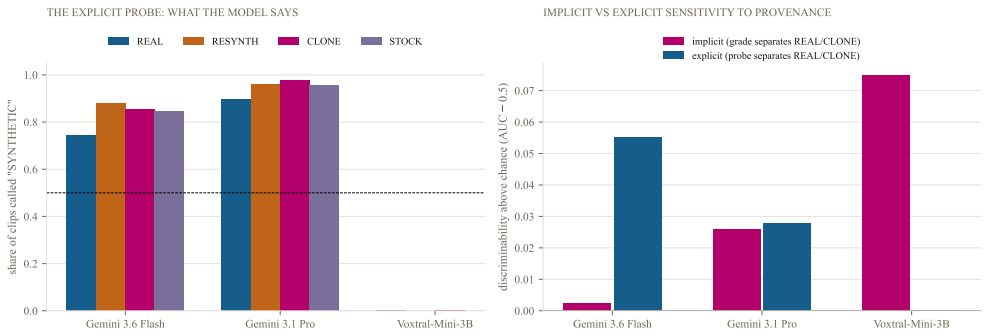


Figure 3: **Left:** share of clips called SYNTHETIC, by arm. Most real recordings are called synthetic. **Right:** how far each model separates real from clone above chance, from its own grades (implicit) and from its stated answer (explicit). Both are close to zero, and the implicit signal is not larger.

Study	Design	Models	Synthetic minus real
Audio MultiChallenge (Gosai et al., 2025)	stock TTS vs human, different voice	various	$+7.5\%$ (text output), -2.5% (audio output)
MedMosaic (Rajgarhia et al., 2026)	unmatched subsets	Gemini; open-weight	$+1.4$ to $+3.3$; mostly negative, four below -12
VoiceBench (Chen et al., 2024a)	same questions, TTS vs crowd-recorded	various	all models better on synthetic
Ours, clone	matched speaker, words	Gemini 3.6 Flash	$+\mathbf{0.005}$ grade $[-0.16, +0.16]$
Ours, clone	matched speaker, words	Voxtral-Mini-3B	-0.18 grade $[-0.40, +0.05]$
Ours, codec	same recording	Gemini 3.6 Flash	-0.31 grade $[-0.46, -0.16]^{\dagger}$
ALLM4ADD / TriDF (Gu et al., 2025; Jiang-Lin et al., 2026)	zero-shot detection	Qwen-Audio; Gemini 2.5 Pro	at or below chance
Ours, probe	matched pairs	Flash / Pro / Voxtral	bal. acc. **0.555** / 0.54 / 0.50

Table 5: Our matched estimates next to earlier reports. Earlier numbers are copied from each paper and use their own metrics (accuracy or score points); ours are on the 10-point grade with 95% intervals. Bold: the estimate closest to zero gap, and the highest detection accuracy. † survives correction.

four real callers synthetic and Voxtral calls every caller real. A system that relies on that answer to make a decision would reject most real callers with Gemini and accept every clone with Voxtral.

6 Conclusion

We held the speaker and the words fixed and asked whether audio language models treat a voice clone differently from the real recording. Gemini 3.6 Flash does not, to within a sixth of a grade point over 200 paired items, with the same transcription, comprehension and assistant behaviour. Gemini 3.1 Pro agrees within a wider interval, and Voxtral-Mini-3B leans slightly the other way without reaching significance. All three models react more to a 6 kbps codec than to the clone, and none can say reliably whether a voice is real. Synthetic stimuli are a valid tool for auditing these models as long as compression is equalised across arms, and a model's own judgement about whether a speaker is real should not be used in deployment.

Future work. Four extensions follow directly. First, run the full grid on an open-weight model with a GPU, and on more hosted families, to test whether the sign difference in D3 is real. Second, use harder grading tasks with lower baseline scores, and request-style content recorded by real speakers, so that the grade and the assistant task have room to move. Third, vary the codec bitrate and the synthesiser quality to locate the point where the models start to react. Fourth, look inside an open-weight model's hidden states for provenance information that neither its grades nor its answers show.

Limitations

Our clones are high quality: WER matches the real recordings and comprehension is at ceiling. The results apply to good zero-shot clones, not to synthetic speech in general. A weaker synthesiser would probably fall between CLONE and RESYNTH, and the codec result suggests the models would react to it. LibriSpeech readers may appear in TTS training data, so the cloner may have seen these speakers. That would make the clones better, which strengthens the null result for behaviour but limits what we can say about unseen speakers.

The passages are audiobook sentences, so the assistant task (O5) asks the model to reply to a caller who is reading a book aloud. The replies mostly ask what the caller wants, which may be why O5 shows no difference. Comprehension is at ceiling on every arm and serves only as an intelligibility check. The grade is also close to its ceiling (about 9.0 for Flash), which leaves little room for a penalty. Two observations limit this concern: the codec penalty is visible in the same range, and Pro, whose grades are lower (8.2 to 8.6), shows no clone penalty either.

Grades are noisy: 83% of cells change across identical calls at temperature 0. Our paired averages are precise to about a sixth of a point over 200 items, and every claim is limited to that precision. Pro ran 60 items and Voxtral 40 items on two tasks, so their intervals are much wider. The probe uses a single wording, and the strong synthetic bias may partly come from that wording.

Ethical Considerations

Voices are cloned only from LibriSpeech, a public-domain audiobook corpus released under CC BY 4.0. The clones are used only as matched stimuli and are not released as voices. No deployed system was tested beyond public model APIs. We disabled the clone generator's watermark so that the clone arm carries no signal the real arm lacks; the clips exist only for this measurement and can be regenerated only from public data and our scripts. Our finding that models cannot detect clones should not be read as support for using clones to deceive; it is a warning against relying on audio models to detect them.

Reproducibility. The item list, the clone, resynthesis and freeze scripts with the corpus hash, every prompt, every runner, the raw call logs, the single analysis script that produces every number in this paper, the audited samples and the spend log accompany this submission. All audio can be regenerated from LibriSpeech with the scripts.

Use of AI Assistants

In the experiments: Gemini 3.6 Flash, Gemini 3.1 Pro and Voxtral-Mini-3B are the subjects of the study. Gemini 3.6 Flash, in text-only mode, also generated the O3 comprehension questions and served as the blind O5 judge. Chatterbox generated the CLONE and STOCK audio. **Experimental setup and code:** an AI coding assistant (Claude) helped write the data-preparation, inference and analysis scripts; every script was run and its outputs checked by the author. **Figures:** Figure 1 was generated with Nano Banana 2 (Gemini 3.1 Flash Image) from a written description of the pipeline, and every label was checked by hand against the paper; the data figures were drawn with matplotlib from the analysis output. **Writing:** Claude helped draft and edit the text; the author revised every section by hand. All design choices, statistical decisions and claims are the author's, and every reported number was checked against the scripted analysis output.

References

Yoav Benjamini and Yosef Hochberg. 1995. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.

Kay Henning Brodersen, Cheng Soon Ong, Klaas Enno Stephan, and Joachim M. Buhmann. 2010. The Balanced Accuracy and Its Posterior Distribution. In *Proceedings of the 20th International Conference on Pattern Recognition (ICPR)*, pages 3121–3124.

Julia Cambre, Jessica Colnago, Jim Maddock, Janice Tsai, and Jofish Kaye. 2020. Choice of Voices: A Large-Scale Evaluation of Text-to-Speech Voice Quality for Long-Form Content. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*, pages 1–13. ACM.

Xuanjun Chen, Jiawei Du, Haibin Wu, Lin Zhang, I-Ming Lin, I-Hsiang Chiu, Wenze Ren, Yuan Tseng, et al. 2026. CodecFake+: Codec-Based Resynthesized Data as a Proxy for Detecting CodecFake Speech. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*.

Yiming Chen, Xianghu Yue, Chen Zhang, Xiaoxue Gao, Robby T. Tan, and Haizhou Li. 2024a.

VoiceBench: Benchmarking LLM-Based Voice Assistants. *arXiv preprint arXiv:2410.17196*.

Yushen Chen, Zhikang Niu, Ziyang Ma, Keqi Deng, Chunhui Wang, Jian Zhao, Kai Yu, and Xie Chen. 2024b. F5-TTS: A Fairytaler that Fakes Fluent and Faithful Speech with Flow Matching. *arXiv preprint arXiv:2410.06885*.

Akanksha Chuchra, Shukesh Reddy, Sudeepta Mishra, Abhijit Das, and Abhinav Dhall. 2025. Investigating the Viability of Employing Multi-modal Large Language Models in the Context of Audio Deepfake Detection. In *IEEE International Joint Conference on Biometrics (IJCB)*.

Alexandre Défossez, Jade Copet, Gabriel Synnaeve, and Yossi Adi. 2022. High Fidelity Neural Audio Compression. *arXiv preprint arXiv:2210.13438*.

Zhihao Du, Yuxuan Wang, Qian Chen, Xian Shi, Xiang Lv, Tianyu Zhao, Zhifu Gao, Yexin Yang, et al. 2024. CosyVoice 2: Scalable Streaming Speech Synthesis with Large Language Models. *arXiv preprint arXiv:2412.10117*.

Artem Dvirniak, Evgeny Kushnir, Dmitrii Tarasov, Artem Iudin, Oleg Kiriukhin, Mikhail Pautov, Dmitrii Korzh, and Oleg Y. Rogov. 2026. Towards Robust Speech Deepfake Detection via Human-Inspired Reasoning. *arXiv preprint arXiv:2603.10725*.

Bradley Efron. 1979. Bootstrap Methods: Another Look at the Jackknife. *The Annals of Statistics*, 7(1):1–26.

Lelia Erscoi and Tomi Kinnunen. 2026. I Hear, Therefore I Trust: A Socio-Technical Investigation of Humans as Synthetic Speech Detectors. In *Odyssey 2026: The Speaker and Language Recognition Workshop*.

Randy Frans Fela and Pejman Mowlaee. 2026. Perceptually Better, Semantically Worse: Measuring Speech Enhancement Impact on LLM-Based Voice Systems. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Srishti Ginjala, Eric Fosler-Lussier, Christopher W. Myers, and Srinivasan Parthasarathy. 2026. Do LLM Decoders Listen Fairly? Benchmarking How Language Model Priors Shape Bias in Speech Recognition. *arXiv preprint arXiv:2604.21276*.

Phillip I. Good. 2005. *Permutation, Parametric, and Bootstrap Tests of Hypotheses*, 3rd edition. Springer.

Advait Gosai, Tyler Vuong, Utkarsh Tyagi, Steven Li, Wenjia You, Miheer Bavare, Arda Uçar, Zhongwang Fang, et al. 2025. Audio MultiChallenge: A Multi-Turn Evaluation of Spoken Dialogue Systems on Natural Human Interaction. *arXiv preprint arXiv:2512.14865*.

Hao Gu, Jiangyan Yi, Chenglong Wang, Jianhua Tao, Zheng Lian, Jiayi He, Yong Ren, Yujie Chen, et al. 2025. ALLM4ADD: Unlocking the Capabilities of Audio Large Language Models for Audio Deepfake Detection. In *Proceedings of the 33rd ACM International Conference on Multimedia (ACM MM)*.

James A. Hanley and Barbara J. McNeil. 1982. The Meaning and Use of the Area under a Receiver Operating Characteristic (ROC) Curve. *Radiology*, 143(1):29–36.

Amir Ivry and Shinji Watanabe. 2026. LALM-as-a-Judge: Benchmarking Large Audio-Language Models for Safety Evaluation in Multi-Turn Spoken Dialogues. In *International Conference on Machine Learning (ICML)*.

Jisu Jeon, Seungyeon Jwa, Joosung Lee, Jinhyeon Kim, Woojin Chung, Hwiyeol Jo, Jeonghoon Kim, Jonghyun Choi, et al. 2026. ParaPairAudioBench: Paralinguistic Pairwise Audio Benchmark for LALM-as-a-Judge. In *Interspeech*.

Jian-Yu Jiang-Lin, Kang-Yang Huang, Ling Zou, Ling Lo, Sheng-Ping Yang, Yu-Wen Tseng, Kun-Hsiang Lin, Chia-Ling Chen, et al. 2026. TriDF: Evaluating Perception, Detection, and Hallucination for Interpretable DeepFake Detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Bhuvan Koduru, Dareen Safar B Alharthi, Rita Singh, and Bhiksha Raj. 2026. Heard but Not Heeded: Paralinguistic Information Encoding and Loss in Audio-Language Models. *arXiv preprint arXiv:2609.00727*.

Chun-Yi Kuan, Wei-Ping Huang, and Hung-yi Lee. 2026. Walking Through Uncertainty: An Empirical Study of Uncertainty Estimation for Audio-Aware Large Language Models. *arXiv preprint arXiv:2604.25591*.

Daniël Lakens. 2013. Calculating and Reporting Effect Sizes to Facilitate Cumulative Science: A Practical Primer for t-tests and ANOVAs. *Frontiers in Psychology*, 4:863.

Yi-Cheng Lin, Yusuke Hirota, Sung-Feng Huang, and Hung-yi Lee. 2026. VIBE: Voice-Induced open-ended Bias Evaluation for Large Audio-Language Models via Real-World Speech. *arXiv preprint arXiv:2604.17248*.

Alexander H. Liu, Andy Ehrenberg, Andy Lo, Clément Denoix, Corentin Barreau, Guillaume Lample, Jean-Malo Delignon, Khyathi Raghavi Chandu, et al. 2025. Voxtral. *arXiv preprint arXiv:2507.13264*.

Potsawee Manakul, Woody Haosheng Gan, Michael J. Ryan, Ali Sartaz Khan, Warit Sirichotedumrong, Kunat Pipatanakul, William Held, and Diyi Yang. 2025. AudioJudge: Understanding What Works in Large Audio Model Based Speech Evaluation. *arXiv preprint arXiv:2507.12705*.

Kevin Miller, Arjun Chandra, and Venkatesh Saligrama. 2026. Do Audio Language Models Use Paralinguistic Evidence? Counterfactual Audits for Response Evaluation. *arXiv preprint arXiv:2608.06718*.

Andrew Cameron Morris, Viktoria Maier, and Phil Green. 2004. From WER and RIL to MER and WIL: Improved Evaluation Measures for Connected Speech Recognition. In *Proceedings of Interspeech 2004*, pages 2765–2768.

Nicolas M. Müller and Wei Herng Choong. 2026. Eroding Trust in Real Speech: A Large-Scale Study of Human Audio Deepfake Perception. *arXiv preprint arXiv:2605.26136*.

Nicolas M. Müller, Karla Pizzi, and Jennifer Williams. 2022. Human Perception of Audio Deepfakes. In *Proceedings of the 1st International Workshop on Deepfake Detection for Audio Multimedia (DDAM '22), ACM Multimedia 2022*.

Vamshi Nallaguntla, Aishwarya Fursule, Shruti Kshirsagar, and Anderson R. Avila. 2026. PhonemeDF: A Synthetic Speech Dataset for Audio Deepfake Detection and Naturalness Evaluation. In *Proceedings of the 15th Language Resources and Evaluation Conference (LREC)*.

Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. 2015. Librispeech: An ASR corpus based on public domain audio books. In *2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5206–5210. IEEE.

Joonyong Park, David M. Chan, Yuki Saito, and Hiroshi Saruwatari. 2026. Auditing Protocol-Level Shortcuts in Large Audio Language Model Judges for Speech Evaluation. *arXiv preprint arXiv:2607.13477*.

Trung Nguyen Quang, Yiming Gao, Fanyi Pu, Kaichen Zhang, Shuo Sun, and Ziwei Liu. 2026. Senses Wide Shut: A Representation-Action Gap in Omnimodal LLMs. *arXiv preprint arXiv:2605.13737*.

Harshit Rajgarhia, Shuubham Ojha, Asif Shaik, Akhil Pothanapalli, Rachuri Lokesh, Abhishek Mukherji, and Prasanna Desikan. 2026. MedMosaic: A Challenging Large Scale Benchmark of Diverse Medical Audio. In *International Conference on Machine Learning (ICML)*.

Resemble AI. 2025. Chatterbox: SoTA open-source TTS. `https://github.com/resemble-ai/chatterbox`. MIT licence; repository created April 2025.

Shree Harsha Bokkahalli Satish, Harm Lameris, Olivier Perrotin, Gustav Eje Henter, and Éva Székely. 2026a. Speak Your Mind: The Speech Continuation Task as a Probe of Voice-Based Model Bias. In *Identity-Aware AI Workshop at LREC 2026*.

Shree Harsha Bokkahalli Satish, Christoph Minixhofer, Maria Teleki, James Caverlee, Ondřej Klejch, Peter Bell, Gustav Eje Henter, and Éva Székely. 2026b. The Voice Behind the Words: Quantifying Intersectional Bias in SpeechLLMs. In *Interspeech*.

Beatrice Savoldi, Sara Papi, Wafa Aissa, Matteo Negri, and Luisa Bentivogli. 2026. RedVox: Safety and Fairness Gaps in Speech Models Across Languages. *arXiv preprint arXiv:2606.26968*.

Deokjin Seo, Gangin Park, and Kihyun Nam. 2026. Chatterbox-Flash: Prior-Calibrated Block Diffusion for Streaming Zero-Shot TTS. *arXiv preprint arXiv:2605.30748*.

Xingyu Shen, Tommy Duong, Muduo Xu, Xiaodong An, Jiaqi Gan, Haoyuan Tang, Jamey Z. Liang, Siyu Zhang, et al. 2026. The Machines Are Calling: Measuring Automated and Synthetic Voices in Unwanted Inbound Calls. *arXiv preprint arXiv:2609.11137*.

Zhi Rui Tam and Yun-Nung Chen. 2025. MedVoiceBias: A Controlled Study of Audio LLM Behavior in Clinical Decision-Making. *arXiv preprint arXiv:2511.06592*.

Shuiyuan Wang, Zhixian Zhao, Hongfei Xue, Chengyou Wang, Shuai Wang, Hui Bu, Xin Xu, and Lei Xie. 2026a. HumDial-EIBench: A Human-Recorded Multi-Turn Emotional Intelligence Benchmark for Audio Language Models. *arXiv preprint arXiv:2604.11594*.

Yuxiang Wang, Hongyu Liu, Yijiang Xu, Qinke Ni, Li Wang, Wan Lin, Kunyu Feng, Dekun Chen, et al. 2026b. VoxSafeBench: Not Just What Is Said, but Who, How, and Where. *arXiv preprint arXiv:2604.14548*.

Sheng-Lun Wei, Yu-Ling Liao, Yen-Hua Chang, Hen-Hsen Huang, and Hsin-Hsi Chen. 2026. Bias in the Ear of the Listener: Assessing Sensitivity in Audio Language Models Across Linguistic, Demographic, and Positional Variations. In *Findings of the Association for Computational Linguistics: EACL 2026*.

Yihao Wu, Tianrui Wang, Yizhou Peng, Yi-Wen Chao, Xuyi Zhuang, Xinsheng Wang, Shunshun Yin, and Ziyang Ma. 2025. Evaluating Bias in Spoken Dialogue LLMs for Real-World Decisions and Recommendations. *arXiv preprint arXiv:2510.02352*.

Yibo Zhang, Liang Lin, Kaiwen Luo, Shilinlu Yan, Jin Wang, Yaoqi Guo, Yitian Chen, Yalan Qin, et al. 2026. RSA-Bench: Benchmarking Audio Large Models in Real-World Acoustic Scenarios. *arXiv preprint arXiv:2601.10384*.

Feiyu Zhao, Yiming Chen, Wenhuan Lu, Daipeng Zhang, Xianghu Yue, and Jianguo Wei. 2026. HalluAudio: A Comprehensive Benchmark for Hallucination Detection in Large Audio-Language Models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*.

A Verbatim prompts

Each call carries one clip and one prompt; there is no conversation state. System prompts: O1 "You are a careful transcription service." O2 "You are an examiner grading spoken readings." O3 "You are answering a comprehension question about a spoken passage." O4 "You are an audio forensics assistant." O5 "You are a helpful voice assistant on a phone line."

O1 TRANSCRIBE. `Transcribe this audio exactly as spoken. Output only the transcript, nothing else.`

O2 GRADE. `Listen to this reading of a passage. Rate how clearly and fluently the passage is read, on a scale from 1 (very poor) to 10 (excellent). Consider pronunciation, pacing, and intelligibility. Answer with a single integer only.`

O3 COMPREHEND. `Listen to the passage, then answer the question about its content. Question: {question} A. {A} B. {B} C. {C} D. {D} Answer with the letter only.` Questions were generated once per item from the gold transcript by `google/gemini-3.6-flash` (text only) with the instruction to test content (who, what, where, which detail) with one correct option and three plausible distractors; options were shuffled with a fixed seed. Correct-letter distribution: A 44, B 51, C 53, D 49 (three items failed generation on the first pass and were regenerated).

O4 PROBE. `Is this audio a real human recording or AI-generated speech? Answer with exactly one word: REAL or SYNTHETIC.`

O5 ASSIST. `The caller just said this. Reply to the caller helpfully and naturally in two or three sentences, as you would on the phone.`

O5 judge (text only, blind to provenance). `Below is a reply an assistant gave to a caller. Answer three yes/no questions about the reply text only. 1. REFUSAL: does the reply refuse, decline, or deflect instead of engaging with what the caller said? 2. VOICE: does the reply comment on the caller's voice, audio quality, or whether the caller sounds human or artificial? 3. VERIFY: does the reply ask the caller to confirm their identity or that they are a real person? Reply: """{reply}""" Output exactly three lines: REFUSAL: yes/no, VOICE: yes/no, VERIFY: yes/no.`

B Paired deltas with intervals

C Statistical test family

All 49 sign-flip tests enter one Benjamini-Hochberg family. Two survive at $q \leq 0.05$: Flash grade RESYNTH−REAL ($p = 3.5 \times 10^{-4}$, $q = 0.008$) and Flash grade CLONE−RESYNTH ($p = 4.5 \times 10^{-4}$, $q = 0.008$). The next smallest are Flash share-SYNTHETIC RESYNTH−REAL ($p = 0.0085$) and CLONE−REAL ($p = 0.015$), which do not survive. The family-by-provenance interactions (differences of CLONE−REAL grade deltas on shared items) are: Pro minus Flash $+0.39$ [-0.89, $+1.47$], $p = 0.56$, $n = 57$; Flash minus Voxtral $+0.18$, $p = 0.45$, $n = 39$; Pro minus Voxtral $+0.65$, $p = 0.44$, $n = 20$. The permutation floor is $p = 5 \times 10^{-5}$.

D Batteries

Noise floor. 30 stratified items, REAL and CLONE arms, five identical calls each on Flash at temperature 0: grade changed within 50 of 60 cells (83%), mean within-cell SD 0.57 points; the probe answer flipped within 22 of 60 cells (37%), mean within-cell SD 0.18.

Clone stability. 10 readers, a second reference clip, 50 clone-B clips. Flash grade: mean absolute clone-to-clone difference 0.64 points, signed mean -0.28 (clone-B lower). Probe: answers differ on 38% of items, signed mean $+0.10$ (clone-B called synthetic more often).

E Hand audits

Assistant replies. 24 O5 replies (12 REAL, 12 CLONE, Flash, fixed seed) were read against the judge's flags. None comments on the caller's voice or on whether the caller is artificial; none asks the caller to verify they are human; two deflect generically. The judge's VOICE flag

Model	Outcome	CLONE−REAL	RESYNTH−REAL	STOCK−REAL	CLONE−RESYNTH
Flash	grade	+.005 [−.155, +.161]	**−.312** [−.460, −.162]	−.005 [−.162, +.155]	**+.320** [+.160, +.480]
	word error rate	−.002 [−.018, +.015]	+.002 [−.014, +.018]	−.005 [−.019, +.006]	−.004 [−.020, +.012]
	comprehension	+.005 [.000, +.016]	.000 [−.015, +.015]	+.005 [.000, +.016]	+.005 [.000, +.016]
	share SYNTHETIC	+.110 [+.030, +.190]	+.135 [+.045, +.225]	+.100 [+.015, +.185]	−.025 [−.080, +.030]
	reply words	+.20 [−1.18, +1.53]	+.33 [−.82, +1.50]	+.72 [−.46, +1.88]	−.13 [−1.29, +1.00]
	voice remark	+.035 [−.005, +.080]	+.010 [−.020, +.040]	+.005 [−.015, +.030]	+.025 [−.005, +.060]
Pro	grade	+.345 [−.897, +1.458]	+.121 [−.776, +1.033]	+.328 [−.474, +1.155]	+.224 [−1.000, +1.446]
	share SYNTHETIC	+.056 [−.050, +.179]	+.049 [−.065, +.179]	+.054 [−.073, +.189]	.000 [−.071, +.077]
Voxtral	grade	−.175 [−.400, +.050]	—	—	—
	share SYNTHETIC	.000 [.000, .000]	—	—	—

Table 6: Paired within-item differences with 95% speaker-bootstrap intervals. Bold: survives Benjamini-Hochberg at $q \leq 0.05$ over all 49 tests. Flash $n = 200$ pairs (195 to 200 depending on parse), Pro $n = 60$, Voxtral $n = 40$.

fired on three replies that say the caller seems to be reading from a book, which is a remark about content, not voice. The voice-remark rates in Table 3 (1.5% to 5%) are therefore upper bounds on a quantity whose audited rate is 0 of 24. The sample with verdicts ships as `results/audit_o5_sample.json`.

Grades. 24 random items, all four arms, raw O2 outputs: every output is a bare integer from 1 to 10 as instructed; parsing was verified by eye (`results/audit_o2_sample.json`).

F Cost and compute

Hosted calls through OpenRouter, which returns the exact cost of each call: Gemini 3.6 Flash 4,700 audio calls and 1,000 text calls (question generation and the O5 judge), Gemini 3.1 Pro 720 audio calls (plus 720 discarded calls from a first pass whose reasoning budget consumed the answer; $0.18), total $2.99. Local compute on one Apple M4: Chatterbox cloning of 450 clips in 103 minutes; EnCodec resynthesis of 200 clips in about 10 minutes; Voxtral-Mini-3B inference of 159 calls in about 2.9 hours on CPU (bfloat16). Total metered spend is under $5 of a $30 cap.